Skill week – 07

WEEK-7:

Task 1: Process Synchronization and Child Process Monitoring Using waitpid().

Task 2: To Retrieve PATH Variable, Parse Search Directories, Locate Executables, Verify Permissions, Handle Missing Commands, Test Command Resolution.

nano task1_waitpid.c

```
#include   <stdio.h>
#include  <stdlib.h>
#include <unistd.h>
#include  <sys/types.h>
#include <sys/wait.h>

 int main() {

    pid_t pid;
   int status;

   printf("Parent process started.\n");
   printf("Parent PID: %d\n\n", getpid());

   pid = fork();

   if (pid < 0) {
      perror("fork failed");
      return 1;
   }

   /* Child process */
   if (pid == 0) {

      printf("Child process started.\n");
      printf("Child PID: %d\n", getpid());
      printf("Child working...\n\n");

      sleep(3);

      printf("Child process finished.\n");

      exit(10);
   }
```

```
    /* Parent process */
    else {

        printf("Parent created child with PID: %d\n", pid);
        printf("Parent is waiting for child...\n\n");

        /*
         * Wait specifically for the child process.
         */
        if (waitpid(pid, &status, 0) == -1) {
            perror("waitpid failed");
            return 1;
        }

        printf("\nParent detected that child has finished.\n");

        if (WIFEXITED(status)) {
            printf("Child exited normally.\n");
            printf("Child exit status: %d\n",
                   WEXITSTATUS(status));
        }

        printf("Parent process finished.\n");
    }

    return 0;
}
```

```
yaswanth6970@Yash6970:~/skill/week07$ nano task1_waitpid.c
yaswanth6970@Yash6970:~/skill/week07$ gcc task1_waitpid.c -o task1
./task1
Parent process started.
Parent PID: 700

Parent created child with PID: 701
Parent is waiting for child...

Child process started.
Child PID: 701
Child working...

Child process finished.

Parent detected that child has finished.
Child exited normally.
Child exit status: 10
Parent process finished.
yaswanth6970@Yash6970:~/skill/week07$
```

nano task2_path.c

```
#include   <stdio.h>
#include  <stdlib.h>
#include  <string.h>
#include <unistd.h>
#include <limits.h>

#define MAX_PATH_LENGTH 4096

/*
 * Search for a command in the directories
 * listed in the PATH environment variable.
 */
void find_command(const char *command) {

   char *path;
   char *path_copy;
   char *directory;

   char full_path[PATH_MAX];

   /* Get PATH environment variable */
   path = getenv("PATH");

   if (path == NULL) {
      printf("PATH variable not found.\n");
      return;
   }

   printf("\nPATH variable:\n");
   printf("%s\n\n", path);

   /* Make a copy because strtok modifies the string */
   path_copy = strdup(path);

   if (path_copy == NULL) {
      perror("strdup  failed");
      return;
   }

   /* Split PATH using ':' */
   directory = strtok(path_copy, ":");
```

```
    while (directory != NULL) {

        /*
         * Construct:
         *
         * directory + "/" + command
         */
        snprintf(full_path,
                sizeof(full_path),
                "%s/%s",
                directory,
                command);

        printf("Checking: %s\n", full_path);

        /*
         * Check whether file exists and
         * has execute permission.
         */
        if (access(full_path, X_OK) == 0) {

            printf("\nFOUND executable!\n");
            printf("Command : %s\n", command);
            printf("Path    : %s\n", full_path);

            free(path_copy);
            return;
        }

        directory = strtok(NULL, ":");
    }

    printf("\nCommand not found: %s\n", command);

    free(path_copy);
}

int main(int argc, char *argv[]) {

    printf("WEEK 7 - TASK 2\n");
    printf("PATH Executable Resolution\n");
    printf("==========================\n");

    /*
     * If the user provides a command:
     *
     * ./task2 ls
```

```
     *
     * use that command.
     */
    if (argc != 2) {

        printf("\nUsage:\n");
        printf("  %s <command>\n\n", argv[0]);

        printf("Examples:\n");
        printf("  %s ls\n", argv[0]);
        printf("  %s gcc\n", argv[0]);
        printf("  %s bash\n", argv[0]);

        return 1;
    }

    printf("\nSearching for command: %s\n", argv[1]);

    find_command(argv[1]);

    return 0;
}
```

```
yaswanth6970@Yash6970:~/skill/week07$ nano task2_path.c
yaswanth6970@Yash6970:~/skill/week07$ gcc -Wall -Wextra -std=gnu11 task2_pat
h.c -o task2
yaswanth6970@Yash6970:~/skill/week07$ ./task2 ls
WEEK 7 - TASK 2
PATH Executable Resolution
==========================

Searching for command: ls

PATH variable:
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr
/local/games:/usr/lib/wsl/lib:/mnt/c/Program Files/WindowsApps/MicrosoftCorp
orationII.WindowsSubsystemForLinux_2.7.11.0_x64__8wekyb3d8bbwe:/mnt/c/Progra
m Files/Common Files/Oracle/Java/javapath:/mnt/c/WINDOWS/system32:/mnt/c/WIN
DOWS:/mnt/c/WINDOWS/System32/Wbem:/mnt/c/WINDOWS/System32/WindowsPowerShell/
v1.0/:/mnt/c/WINDOWS/System32/OpenSSH/:/mnt/c/Program Files/NVIDIA Corporati
on/NVIDIA App/NvDLISR:/mnt/c/Program Files/Git/cmd:/mnt/c/Program Files/node
js/:/mnt/c/Program Files (x86)/VMware/VMware Workstation/bin/:/mnt/c/Program
 Files/Common Files/Oracle/Java/javapath:/mnt/c/WINDOWS/system32:/mnt/c/WIND
OWS:/mnt/c/WINDOWS/System32/Wbem:/mnt/c/WINDOWS/System32/WindowsPowerShell/v
1.0/:/mnt/c/WINDOWS/System32/OpenSSH/:/mnt/c/Program Files/NVIDIA Corporatio
n/NVIDIA App/NvDLISR:/mnt/c/Program Files/Git/cmd:/mnt/c/Program Files/MySQL
/MySQL Shell 8.0/bin/:/mnt/c/Users/chyas/AppData/Local/Microsoft/WindowsApps
:/mnt/c/Users/chyas/AppData/Local/Programs/Microsoft VS Code/bin:/mnt/c/Prog
ram Files/Java/jdk-21.0.12/bin:/mnt/c/Users/chyas/AppData/Local/Python/bin:/
mnt/c/Users/chyas/AppData/Roaming/npm:/snap/bin

Checking: /usr/local/sbin/ls
Checking: /usr/local/bin/ls
Checking: /usr/sbin/ls
Checking: /usr/bin/ls

FOUND executable!
Command : ls
Path    : /usr/bin/ls
```

```
yaswanth6970@Yash6970:~/skill/week07$ ./task2 gcc
WEEK 7 - TASK 2
PATH Executable Resolution
==========================

Searching for command: gcc

PATH variable:
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr
/local/games:/usr/lib/wsl/lib:/mnt/c/Program Files/WindowsApps/MicrosoftCorp
orationII.WindowsSubsystemForLinux_2.7.11.0_x64__8wekyb3d8bbwe:/mnt/c/Progra
m Files/Common Files/Oracle/Java/javapath:/mnt/c/WINDOWS/system32:/mnt/c/WIN
DOWS:/mnt/c/WINDOWS/System32/Wbem:/mnt/c/WINDOWS/System32/WindowsPowerShell/
v1.0/:/mnt/c/WINDOWS/System32/OpenSSH/:/mnt/c/Program Files/NVIDIA Corporati
on/NVIDIA App/NvDLISR:/mnt/c/Program Files/Git/cmd:/mnt/c/Program Files/node
js/:/mnt/c/Program Files (x86)/VMware/VMware Workstation/bin/:/mnt/c/Program
 Files/Common Files/Oracle/Java/javapath:/mnt/c/WINDOWS/system32:/mnt/c/WIND
OWS:/mnt/c/WINDOWS/System32/Wbem:/mnt/c/WINDOWS/System32/WindowsPowerShell/v
1.0/:/mnt/c/WINDOWS/System32/OpenSSH/:/mnt/c/Program Files/NVIDIA Corporatio
n/NVIDIA App/NvDLISR:/mnt/c/Program Files/Git/cmd:/mnt/c/Program Files/MySQL
/MySQL Shell 8.0/bin/:/mnt/c/Users/chyas/AppData/Local/Microsoft/WindowsApps
:/mnt/c/Users/chyas/AppData/Local/Programs/Microsoft VS Code/bin:/mnt/c/Prog
ram Files/Java/jdk-21.0.12/bin:/mnt/c/Users/chyas/AppData/Local/Python/bin:/
mnt/c/Users/chyas/AppData/Roaming/npm:/snap/bin

Checking: /usr/local/sbin/gcc
Checking: /usr/local/bin/gcc
Checking: /usr/sbin/gcc
Checking: /usr/bin/gcc

FOUND executable!
Command : gcc
Path    : /usr/bin/gcc
yaswanth6970@Yash6970:~/skill/week07$ ./task2 bash
WEEK 7 - TASK 2
PATH Executable Resolution
==========================

Searching for command: bash

PATH variable:
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr
/local/games:/usr/lib/wsl/lib:/mnt/c/Program Files/WindowsApps/MicrosoftCorp
orationII.WindowsSubsystemForLinux_2.7.11.0_x64__8wekyb3d8bbwe:/mnt/c/Progra
m Files/Common Files/Oracle/Java/javapath:/mnt/c/WINDOWS/system32:/mnt/c/WIN
DOWS:/mnt/c/WINDOWS/System32/Wbem:/mnt/c/WINDOWS/System32/WindowsPowerShell/
v1.0/:/mnt/c/WINDOWS/System32/OpenSSH/:/mnt/c/Program Files/NVIDIA Corporati
on/NVIDIA App/NvDLISR:/mnt/c/Program Files/Git/cmd:/mnt/c/Program Files/node
js/:/mnt/c/Program Files (x86)/VMware/VMware Workstation/bin/:/mnt/c/Program
 Files/Common Files/Oracle/Java/javapath:/mnt/c/WINDOWS/system32:/mnt/c/WIND
OWS:/mnt/c/WINDOWS/System32/Wbem:/mnt/c/WINDOWS/System32/WindowsPowerShell/v
1.0/:/mnt/c/WINDOWS/System32/OpenSSH/:/mnt/c/Program Files/NVIDIA Corporatio
n/NVIDIA App/NvDLISR:/mnt/c/Program Files/Git/cmd:/mnt/c/Program Files/MySQL
/MySQL Shell 8.0/bin/:/mnt/c/Users/chyas/AppData/Local/Microsoft/WindowsApps
:/mnt/c/Users/chyas/AppData/Local/Programs/Microsoft VS Code/bin:/mnt/c/Prog
ram Files/Java/jdk-21.0.12/bin:/mnt/c/Users/chyas/AppData/Local/Python/bin:/
mnt/c/Users/chyas/AppData/Roaming/npm:/snap/bin

Checking: /usr/local/sbin/bash
Checking: /usr/local/bin/bash
Checking: /usr/sbin/bash
Checking: /usr/bin/bash

FOUND executable!
Command : bash
Path    : /usr/bin/bash
```

```
yaswanth6970@Yash6970:~/skill/week07$ ./task2 xyzcommand123
WEEK 7 - TASK 2
PATH Executable Resolution
==========================

Searching for command: xyzcommand123

PATH variable:
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/wsl/lib:/mnt/c/P
rogram Files/WindowsApps/MicrosoftCorporationII.WindowsSubsystemForLinux_2.7.11.0_x64__8wekyb3d8bbwe:/mnt/c/Progra
m Files/Common Files/Oracle/Java/javapath:/mnt/c/WINDOWS/system32:/mnt/c/WINDOWS:/mnt/c/WINDOWS/System32/Wbem:/mnt
/c/WINDOWS/System32/WindowsPowerShell/v1.0/:/mnt/c/WINDOWS/System32/OpenSSH/:/mnt/c/Program Files/NVIDIA Corporati
on/NVIDIA App/NvDLISR:/mnt/c/Program Files/Git/cmd:/mnt/c/Program Files/nodejs/:/mnt/c/Program Files (x86)/VMware/
VMware Workstation/bin/:/mnt/c/Program Files/Common Files/Oracle/Java/javapath:/mnt/c/WINDOWS/system32:/mnt/c/WIND
OWS:/mnt/c/WINDOWS/System32/Wbem:/mnt/c/WINDOWS/System32/WindowsPowerShell/v1.0/:/mnt/c/WINDOWS/System32/OpenSSH/:
/mnt/c/Program Files/NVIDIA Corporation/NVIDIA App/NvDLISR:/mnt/c/Program Files/Git/cmd:/mnt/c/Program Files/MySQL
/MySQL Shell 8.0/bin/:/mnt/c/Users/chyas/AppData/Local/Microsoft/WindowsApps:/mnt/c/Users/chyas/AppData/Local/Prog
rams/Microsoft VS Code/bin:/mnt/c/Program Files/Java/jdk-21.0.12/bin:/mnt/c/Users/chyas/AppData/Local/Python/bin:/
mnt/c/Users/chyas/AppData/Roaming/npm:/snap/bin

Checking: /usr/local/sbin/xyzcommand123
Checking: /usr/local/bin/xyzcommand123
Checking: /usr/sbin/xyzcommand123
Checking: /usr/bin/xyzcommand123
Checking: /sbin/xyzcommand123
Checking: /bin/xyzcommand123
Checking: /usr/games/xyzcommand123
Checking: /usr/local/games/xyzcommand123
Checking: /usr/lib/wsl/lib/xyzcommand123
Checking: /mnt/c/Program Files/WindowsApps/MicrosoftCorporationII.WindowsSubsystemForLinux_2.7.11.0_x64__8wekyb3d8
bbwe/xyzcommand123
Checking: /mnt/c/Program Files/Common Files/Oracle/Java/javapath/xyzcommand123
Checking: /mnt/c/WINDOWS/system32/xyzcommand123
Checking: /mnt/c/WINDOWS/xyzcommand123
Checking: /mnt/c/WINDOWS/System32/Wbem/xyzcommand123
Checking: /mnt/c/WINDOWS/System32/WindowsPowerShell/v1.0//xyzcommand123
Checking: /mnt/c/WINDOWS/System32/OpenSSH//xyzcommand123
Checking: /mnt/c/Program Files/NVIDIA Corporation/NVIDIA App/NvDLISR/xyzcommand123
Checking: /mnt/c/Program Files/Git/cmd/xyzcommand123
Checking: /mnt/c/Program Files/nodejs//xyzcommand123
Checking: /mnt/c/Program Files (x86)/VMware/VMware Workstation/bin//xyzcommand123
Checking: /mnt/c/Program Files/Common Files/Oracle/Java/javapath/xyzcommand123
Checking: /mnt/c/WINDOWS/system32/xyzcommand123
Checking: /mnt/c/WINDOWS/xyzcommand123
Checking: /mnt/c/WINDOWS/System32/Wbem/xyzcommand123
Checking: /mnt/c/WINDOWS/System32/WindowsPowerShell/v1.0//xyzcommand123
Checking: /mnt/c/WINDOWS/System32/OpenSSH//xyzcommand123
Checking: /mnt/c/Program Files/NVIDIA Corporation/NVIDIA App/NvDLISR/xyzcommand123
Checking: /mnt/c/Program Files/Git/cmd/xyzcommand123
Checking: /mnt/c/Program Files/MySQL/MySQL Shell 8.0/bin//xyzcommand123
Checking: /mnt/c/Users/chyas/AppData/Local/Microsoft/WindowsApps/xyzcommand123
Checking: /mnt/c/Users/chyas/AppData/Local/Programs/Microsoft VS Code/bin/xyzcommand123
Checking: /mnt/c/Program Files/Java/jdk-21.0.12/bin/xyzcommand123
Checking: /mnt/c/Users/chyas/AppData/Local/Python/bin/xyzcommand123
Checking: /mnt/c/Users/chyas/AppData/Roaming/npm/xyzcommand123
Checking: /snap/bin/xyzcommand123

Command not found: xyzcommand123
yaswanth6970@Yash6970:~/skill/week07$
```